TRANSPORTED ANTI-SPOOFING THRESHOLDS CAN FAIL CONSERVATIVELY: MISSED SPOOFS AT A FALSE-ALARM RATE BELOW TARGET

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ABSTRACT

A speech-deepfake detector's threshold, set to 5% false-positive rate (FPR, bona fide rejected) on one channel, often loses that operating point on the channels and corpora tested here. Across 108 detector, source and target combinations built from the real transmissions of ASVspoof 2021 LA and four corpora, 77 realize an FPR more than 2× off target (36 above, 41 below). Of the 72 cells within our own ±5-percentage-point FPR tolerance, 41 fall below half the target: the tolerance reads clean where spoofs pass. In one AASIST example, a threshold calibrated on a PSTN transmission and deployed on the same recordings over G.722 realizes **0.42% FPR** while missing **70% of spoofs, against 0.66% at the deployment-calibrated threshold**. The failure persists on ASVspoof 5 when calibration and deployment share neither recordings nor speakers: 187 of 264 ordered pairs (71%) miss by more than 2× across two detectors, and on the 66 SSL-AASIST pairs whose destinations keep oracle FNR at or below 50% for every source, the transported threshold misses a median 41.5 points more spoofs than the deployment-calibrated one. A target-channel bona-fide order statistic restores *marginal expected* FPR control: at N=500, mean realized FPR is 4.91–5.06% over twelve detector–corpus cells, with mean false-negative rate (FNR) at most 0.5 points above a threshold fitted on all deployment bona fide in the run shown, in cells where that reference misses at most 50% of spoofs. The guarantee is marginal, not per deployment: its Beta dispersion law assumes iid sampling, and speaker clustering widens it on every tested cell. An importance-weighted quantile without target labels does not reliably restore near-target FPR; a mixture-distance drift monitor is defeated by attack-prevalence shifts, and an entropy monitor never flags drifted cells at TPR≥0.8 with FPR≤0.2. Threshold transport can therefore fail quietly, and target-channel calibration must report its sampling unit and dispersion.

Index Terms**—** Anti-spoofing, audio deepfake detection, threshold calibration, conformal prediction, domain shift

1. INTRODUCTION

Anti-spoofing studies usually report EER, an oracle operating point chosen with labels from both classes; a deployed service must fix a threshold in advance and accept the resulting false-positive rate. A threshold set to 5% FPR on ASVspoof 2019 LA dev [1] realizes 22–98.5% FPR across 21LA/21DF [2], In-the-Wild [3], and BRSpeech-DF [4] for a strong open-source detector [5]. Transfer failures of this kind are documented in [6, 7], and presentation can matter more than model scale [8]. LRLspoof [9] transfers an EER threshold across 66 languages but, without target-domain bona fide, can report only spoof rejection; our regime assumes the converse resource, target bona fide without target spoof labels, so we evaluate the FPR side.

We study the deployment behavior of a standard statistical repair: given N labeled target-domain *bona-fide* recordings and no spoof labels, the $\lfloor (N+1)\alpha \rfloor$-th order statistic has marginal rank validity under exchangeability without ties, and under iid continuous sampling its conditional FPR follows an exact finite-sample Beta law [10, 11]; conservative Neyman–Pearson variants give high-probability upper bounds [10]. Neither the statistic nor bona-fide-only audio calibration is new [12]. Our contribution is empirical: we measure transport across channels and corpora, its spoof-side cost, its persistence on disjoint data, and what alternative policies buy at the same labeling budget.

We make three contributions (details in §§4–5). (1) The measurement covers 108 deployment cells: two detectors × (42 ordered pairs of seven 21LA channel conditions + 12 ordered pairs of four corpora), reporting both the loss of false-alarm control and the spoof-side cost relative to each deployment's oracle. (2) We compare calibration policies at matched labeling resource. (3) We report negative results for a label-free heuristic and two unlabeled drift monitors.

2. RELATED WORK

Order-statistic calibration is established: Neyman–Pearson methods control type-I error through class-0 order statistics [10] and negative-only conformal calibration provides marginal guarantees [11]. In audio, CA-SOADD calibrates a bona-fide-only α-quantile [12], [6] audits threshold transfer and unlabeled corrections, and [7] documents bona-fide-driven cross-testing errors. Relative to that audit (one detector, two target corpora), we add a 108-cell fixed-FPR map with real transmissions, a finite-N target-bona-fide comparison, and a 264-pair disjoint replication. Auditing both error rates at a preset threshold is standard biometric practice [13, 14, 15], and [16] holds a cost-optimal threshold fixed across the same 21LA codecs but reports relative cost only; what remains open is how a threshold set on one channel behaves on another when only bona fide is available to reset it, which is what we measure. Neighbouring work examines threshold transfer for partial deepfakes [17], under bona-fide resource shifts [18], and in an industry report without reproducible figures [19], label-free speaker sensitivity [20], and calibration from labeled examples of both classes [21]; our question is finite-sample reset from target bona fide alone. ReplayDF [22] reports channel-induced spoof-side degradation with stable bona-fide acceptance; we measure it at a fixed bona-fide error target.

ASVspoof 5's standalone track reports minDCF, actDCF, C_{llr} and EER [15], and tandem evaluation uses t-DCF [14]; we report EER and both class-conditional errors at a prescribed 5% bona-fide rejection target because our question is threshold control, not an application-cost mixture.

A pre-specified contamination test, replacing 25 of the 500 cohort recordings with spoofs, pushes realized FPR below target on seven of eight cells (0.31–4.96%); the eighth, overlap-dominated cell stays at 5.21%. We test only the score-weighting heuristic of §4, so its negative result is not attributed to covariate-shift weighting [23] or adaptive conformal inference [24].

3. THRESHOLD POLICIES

We treat spoof as the positive class. A detector emits score s (higher = bona fide), and deployment flags spoof when $s < t$, so $\mathrm{FPR}(t) = \Pr(s < t \mid \text{bona fide})$ is the bona-fide rejection rate ($P_{\mathrm{miss}}^{\mathrm{cm}}$ in the ASVspoof convention) and $\mathrm{FNR}(t) = \Pr(s \geq t \mid \text{spoof})$ the spoof acceptance rate. "Calibration" below means selecting t for a target FPR, not probability calibration. We compare policies by resource class. With **no target data**, the source dev threshold t at the target FPR is transferred unchanged. With **unlabeled target data**, we fit variants of the corrections of [6]: each transform is fitted to the unlabeled source and target mixtures separately, the 5% quantile of transformed source bona fide is taken, and that threshold is applied to transformed target scores. C1 is z-norm; C2 pseudo-labels the top and bottom score quartiles and fits a logistic transform by 50 damped Newton steps from w=1, b=0 (each step halved until the loss decreases; $10^{-6}I$ on the Hessian), whose endpoint on these separable pseudo-labels defines the transform; C5 is AS-norm over the 100 nearest of at most 5,000 cohort embeddings, without self-exclusion. With N **labeled target bona-fide recordings**, we test three policies: (a) cohort z-norm standardizes scores by the cohort mean and standard deviation, then applies the source threshold in standardized units; (b) the Gaussian parametric quantile sets $t = \bar{s}_N - 1.645\,\hat{\sigma}_N$; and (c) the *conformal quantile* sets t to the k-th smallest cohort score, where $k = \lfloor (N{+}1)\alpha \rfloor$. Exchangeability without ties gives the last policy's marginal rank guarantee. Under iid continuous sampling from a fixed score distribution, its conditional FPR across calibration draws additionally follows $\mathrm{Beta}(k, N{+}1{-}k)$ [10, 11]. The **target-bona-fide oracle** sets the threshold realizing α on deployment bona fide; spoof labels measure its FNR but do not select it. **Definitions.** A detector–corpus cell is *overlap-dominated* when its oracle FNR at the target FPR exceeds 50%: it has no usable operating point under our joint criterion, is italicised in Table 1, and is excluded from claims about usable cells. The Beta dispersion is an iid-continuous population reference, not the law of finite-pool draws without replacement; we report the empirical spread separately. C4 (CORAL) of [6] additionally requires applying the frozen classifier head to transformed embeddings and is not evaluated.

4. EXPERIMENTS

Setup. We evaluate AASIST [25], SSL-AASIST [5], and XLS-R+SLS [26]. The first two use released 19LA-trained checkpoints; where official XLS-R+SLS scores are available, our scores agree at per-trial r=0.994. The corpora are ASVspoof 2021 LA and DF [2], In-the-Wild [3], BRSpeech-DF [4], and ASVspoof 5 eval [15] for replication. The 2021 keys carry *eval*, *progress* and *hidden* phases; the hidden phase removes non-speech by voice-activity detection and was not used in challenge results [27]. Non-speech is a documented shortcut for these detectors [16, 28], so the trimmed subset is a shortcut-removed benchmark [29] rather than a deployment channel, and mixing it into a calibration cohort mixes two score populations. A first version of this study did so; as a post-hoc protocol correction, every result below excludes it and keeps the two untrimmed phases, and restricting to the eval phase alone changes no count reported below. Each 21LA condition is then the same 2,356 bona-fide recordings of 67 speakers and 21,164 spoofs: six conditions are real transmissions, five through an Asterisk PBX over VoIP and one over a PSTN carrier, and one is the untransmitted source [2]. By contrast, 21DF is compression-only and untransmitted. Every detector is evaluated on the full 21LA and 21DF sets (16,492 and 20,637 bona fide; five 21DF spoof files were undecodable for our two detectors and excluded). ASVspoof 5 has no hidden phase. For cross-corpus transport the 21LA node is its untransmitted condition only (2,356 bona-fide and 21,164 spoof trials); Table 1 instead pools all seven conditions. All thresholds target α=5%. Within each Monte Carlo run, labeled policies share paired cohort draws (B=1000); Table 1 combines unpaired means from separate runs. FPR is measured on held-out data.

FPR control and FNR cost. At N=500, the conformal quantile's mean realized FPR is 4.91–5.06% across the 12 detector–corpus cells (Table 1), and in a separate 1000-draw diagnostic on the same score sets the 2.5–97.5 percentile spread of realized FPR lies within 2.8–7.6% on every cell, close to the iid Beta reference. On every cell with a usable operating point, the quantile's mean FNR premium over the oracle threshold is $\leq$0.5 points.

Matched-resource policy comparison. The lower block of Table 1 compares the policies of §3 on SSL-AASIST at N=500 over 1000 draws. The conformal quantile realizes 4.9–5.0% FPR on all four corpora, as marginal rank validity predicts. Naive transfer and the unlabeled corrections, as implemented, miss the target by up to 95 pp. The Gaussian parametric quantile is the fair competitor at the same N because it also targets a target-domain FPR; it misses by up to 6.9 pp and collapses to 0% on BRSpeech. Cohort z-norm does not target a target-domain FPR, so its larger errors (6/8 detector–corpus cells beyond $\pm$2 pp, up to 50.8 pp for XLS-R+SLS) are a category difference rather than a direct contest.

Drift is detector-conditioned (Fig. 1). A cell reports the mean deployment FPR over 1000 thresholds, each set from 500 source bona-fide recordings. Its severity is the $\log_2$ ratio of $\max(\mathrm{FPR}, 3/n)$ to α, with n the deployment bona-fide count. Our $\pm$5-point band around a 5% target includes zero by construction, so it cannot distinguish a collapsed threshold from a controlled one; an absolute tolerance narrower than the target would not share this defect. Over the 108 cells, 77 miss target by more than 2$\times$ (57 of the 84 within-corpus cells). On the ratio scale, 41 of the 72 cells inside our own $\pm$5 pp tolerance on realized FPR miss target by over 2$\times$, necessarily conservatively, since that tolerance excludes a liberal 2$\times$ miss by construction. The 2$\times$ bar is post-hoc, and the count depends on that choice: it ranges from 54 of 72 at 1.5$\times$ to 29 of 72 at 4$\times$. We therefore attach no interval. 58 of those 72 cells are within-corpus, reordering the same 2,356 recordings of 67 speakers, so they are not independent observations and no significance claim is made of them; the replicate below tests whether the finding survives this dependence. Their spoof-side cost is measured against the oracle threshold for that deployment, not against the calibration condition, and reaches +69.4 percentage points (AASIST, PSTN$\rightarrow$G.722, whose 0.42% FPR is about 10 false alarms per draw among 2,356 bona fide; two cells score marginally higher but their FPR rests on fewer than five expected false alarms, $\mathrm{FPR} < 5/n$, and is censored). Every AASIST cell calibrated on PSTN pays at least +40 points; the same threshold misses 31% of PSTN spoofs, and PSTN is AASIST's least separable condition (AUC 0.956 against 0.975–0.997), so its threshold sits below the bona-fide scores of cleaner channels. A positive spoof-side cost places the threshold below oracle and so inside the $\pm$5 pp tolerance, which is why the additive test cannot reveal it; empirically every such cell passes. **Replication on recording-disjoint data.** Because the grid above reorders one 2,356-recording set, we repeat the experiment on ASVspoof 5 eval [15], where calibration and deployment share neither recording nor speaker. The grid contains 11 organizer-applied codecs plus the unprocessed source and 737 speakers. It is restricted to the 80.5% of bona-fide sources appearing under one condition and split 50/50 by speaker. **Over both detectors the transported threshold misses target by over 2$\times$ on 187 of 264 pairs (71%), compared with 57 of 84 within-21LA pairs (68%)**: the effect persists without shared recordings or speakers; the numerical rate difference is confounded by corpus, codecs and attacks. **The spoof-side replication is narrower.** A no-codec pilot with 1,500 recordings per class gave oracle FNRs of 63.9% (AASIST) and 7.9% (SSL-AASIST); in the disjoint evaluation every AASIST destination and six of the twelve SSL-AASIST destinations exceed 50% oracle FNR. Spoof-side conclusions are therefore restricted to SSL-AASIST's other six destinations (66 ordered pairs), where 49 miss the FPR target conservatively by more than 2$\times$ and 51 pay a positive spoof-side cost; across all 66 pairs the median cost is +41.5 points and 47 exceed +10 (Fig. 1, filled triangles). A same-condition control, calibrating on one speaker half and deploying on the other under the same codec, realizes 3.2–6.0% FPR (SSL-AASIST) and 3.8–7.3% (AASIST) over the twelve conditions with no 2$\times$ miss, so the speaker split alone produces no 2$\times$ miss in these same-condition controls. The flagship AASIST cell does not replicate; the spoof-side axis here rests on one detector. On this deployment half (34,539 bona fide, 312,440 spoofs) the pooled EER is 30.6% for AASIST and 11.8% for SSL-AASIST, not a full-protocol value. On three further detectors with released 21LA scores (XLS-R+SLS, XLSR-Mamba [30], and the fixed-size XLSR-Conformer [31]), the transported threshold misses target by more than 2$\times$ on 25–33 of the 42 channel pairs each (88 of 126), while their spoof-side cost on PSTN$\rightarrow$G.722 stays below 3 points: the FPR-axis failure appears on all five tested detectors, the spoof-side magnitude does not.

Monitoring. We evaluated two unlabeled drift monitors, pre-specified with the acceptance criterion TPR $\geq$ 0.8 and FPR $\leq$ 0.2 for flagging drifted cells. The flag definition was changed after results existed, from "realized FPR more than 2 pp off target" to $|\log_2(\mathrm{FPR}/\alpha)| > 1$; under the pre-specified definition no monitor has an operating point. Under the post-hoc alternative definition the mixture-W_1 monitor, the W_1 distance between deployment and calibration mixture-score distributions, has an in-sample operating point on SSL-AASIST only (threshold selected on the same cells), but multiplying the deployment spoof-to-bona-fide odds by 0.5 and 1.5 with the bona-fide set fixed pushes 19/19 and 18/19 benign cells over it. The entropy monitor, the mean binary

Table 1. Upper block: mean FNR (%) over B=1000 paired N=500 cohorts at the conformal quantile targeting FPR=5%; parentheses give mean FNR minus deployment-oracle FNR, in points. Per-condition 21LA EER ranges: 1.9–11.1 (AASIST), 0.2–1.0 (SSL-AASIST), 0.5–3.5 (XLS-R+SLS). Lower block: mean realized FPR of each policy of §3 for SSL-AASIST at the same N and B from separate runs (naive transfer and C1/C2/C5 are deterministic); C5 reaches its FPR only at FNR $\approx$99%. Italics: overlap-dominated cells (oracle FNR >50%), where FPR control holds but no usable operating point exists. Code, score tables and results (input roots via environment variables; C5 also needs the cohort embeddings, not included): `https://github.com/rvirgilli/speech-deepfake-threshold-transport/tree/a2-icassp2027-final`.

System	21LA	21DF	ITW	BRSpeech
AASIST	19.1 (+0.4)	36.7 (+0.1)	*95.2 (−0.1)*	*84.0 (−0.3)*
SSL-AASIST	0.2 (+0.0)	6.9 (+0.1)	18.3 (+0.2)	*98.5 (0.0)*
XLS-R+SLS†	1.5 (+0.1)	0.7 (+0.1)	11.3 (+0.5)	*91.5 (+0.4)*
EER (%) on the same trials				
AASIST	8.3	14.2	43.0	36.1
SSL-AASIST	0.8	5.9	11.2	34.7
XLS-R+SLS†	2.8	2.0	7.5	34.2
Realized FPR (%) per policy, SSL-AASIST, N=500				
Naive transfer	22.2	38.4	98.5	95.7
C1 z-norm	18.7	24.5	100.0	100.0
C2 temp./shift	100.0	100.0	100.0	100.0
C5 AS-norm	1.4	6.4	0.0	0.1
Cohort z-norm	7.4	20.3	11.9	23.2
Gaussian q.	5.6	11.9	5.7	0.0
Conformal q.	5.0	5.0	5.0	4.9

† Official author-released scores for 21LA, 21DF and ITW; BRSpeech scored by us with the released checkpoint.

entropy of $p_i = \mathrm{clip}(\sigma((s_i - \mu)/\varsigma), 10^{-6}, 1 - 10^{-6})$ over deployment scores with μ, ς the calibration-mixture mean and standard deviation, alerting on large values, fails on both detectors under either definition. **Label-free heuristic.** We test an importance-weighted quantile that reweights the 500 labeled source-domain bona-fide scores of the calibration cohort by a clipped ratio of unlabeled deployment to calibration *mixture-score* densities (15 equal-width bins over the pooled score range, ratio clipped to $[0.1, 10]$, 200 draws); it uses no target labels. It places 17/108 cells within 2 pp of the FPR target, compared with 20/108 unweighted, so it does not reliably restore near-target FPR across this grid; we did not evaluate its spoof-side error. An adaptive-conformal FPR update [24] needs labeled bona-fide outcome feedback, which the unlabeled flag rate does not identify when attack prevalence is unknown.

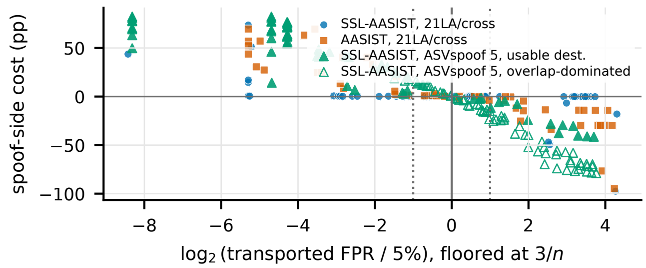

Fig. 1. Transported FPR against spoof-side cost. Each point is one deployment cell: severity $\log_2(\max(\mathrm{FPR}, 3/n)/5\%)$ (§4; dotted lines mark the 2× bar) against mean FNR at the transported threshold minus FNR at the deployment-calibrated threshold, in points. Circles and squares are the 108 21LA and cross-corpus cells; triangles are the 132 ASVspoof 5 SSL-AASIST pairs, filled where the destination is usable (oracle FNR $\leq$50% for every source) and hollow where it is overlap-dominated. Conservative misses (left of centre) carry the positive costs.

5. DISCUSSION

Calibration and detection are distinct. Target-channel calibration controls the *marginal mean*, not every realized threshold. Under iid continuous sampling from a fixed distribution, a single N=100 draw has only 65.5% probability of landing within ±2 FPR points; at N=500, the probability is 96.2%, with a 3.26–7.06% central 95% interval. **Operational implications.** Report FNR at controlled FPR alongside EER, choose N from the dispersion required by the deployment, and recalibrate after a channel change. **Limitations.** A post-hoc speaker-disjoint diagnostic on 21LA, which calibrates on whole speakers until the cohort holds at least 500 recordings (median 15 speakers) and evaluates on the rest, widened the 5th–95th percentile spread of realized FPR on all six tested detector–condition cells (AASIST and SSL-AASIST on the untransmitted, PSTN and GSM conditions) to 6.0–10.1 points (means over ten seeds of 400 draws), against a median of 3.4–3.5 under a null that reassigns individual bona-fide scores to fixed speaker-sized groups (200 permutations); deployment to new speakers therefore requires speaker-level validation, and speaker identity is itself a documented source of detector variation [32]. Conclusions rest on three detectors (the drift map and replicate on two) and one language-shift corpus (n=1), and assume bona-fide labeling is cheap relative to spoof labeling, as when a deployment can verify its own recordings.

6. CONCLUSION

Threshold transport is easy to mis-audit: conservative collapse makes FPR read below target while missed spoofs multiply. A target-channel bona-fide order statistic restores marginal expected control; its dispersion depends on the sampling unit, and it cannot repair class overlap.

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